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Example 8.5. a^n can be defined recursively as

$$a^n = \begin{cases} 1 & \text{if } n = 0 \\ a \times a^{n-1} & \text{otherwise} \end{cases}$$

The recursive definition can be expressed as a recursive solver for computing power(a, n).

```
power (a, n)

-- inputs: n is an integer , n ≥ 0

-- outputs : a^n

if n = 0 -- base case

  1

else    --recursion step

        a × power (a, n-1)
```

The recursive process with solvers for calculating power(2, 5) is shown in Figure 8.8.

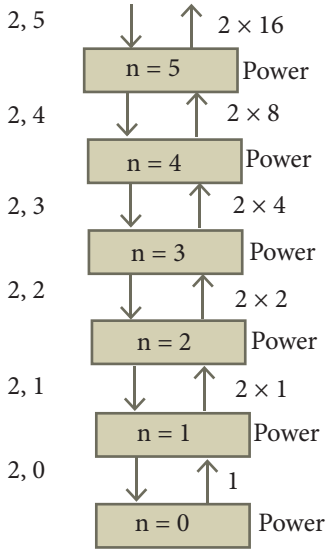

Figure 8.8: Recursive process with solvers for calculating power(2, 5)

The recursive process resulting from power(2, 5) is shown in Figure 8.9.

```
power (2,5)
= 2 × power (2,4)
= 2 × 2 × power(2,3)
= 2 × 2 × 2 × power(2, 2)
= 2 × 2 × 2 × 2 × power (2,1)
= 2 × 2 × 2 × 2 × 2 × power (2,0)
= 2 × 2 × 2 × 2 × 2 × 1
= 2 × 2 × 2 × 2 × 2
= 2 × 2 × 2 × 4
= 2 × 2 × 8
= 2 × 16
= 32
```

Figure 8.9: Recursive process for power(2, 5)

Example 8.13. A corner-covered board is a board of $2^n \times 2^n$ squares in which the square at one corner is covered with a single square tile. A triominoe is a L-shaped tile formed with three adjacent squares (see Figure 8.10). Cover the corner-covered board with the L-shaped triominoes without overlap. Triominoes can be rotated as needed.

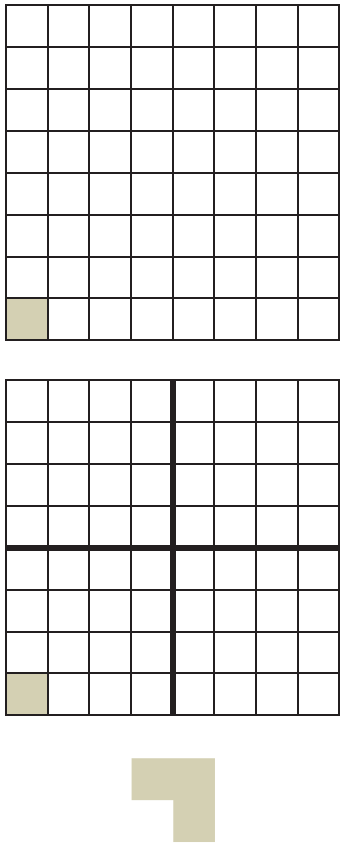

Figure 8.10: Corner-covered board and triominoe